



Research on the impact of digital finance on the high-quality development of education

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ABSTRACT

Against the backdrop of building a strong nation in education, the advantages of digital finance become increasingly prominent, providing motivation for the high-quality development of the education industry. This article utilizes provincial panel data from China covering the period from 2011 to 2022 to construct an index system for high-quality education development and conduct regression estimation. The results indicate that digital finance plays a significant promoting role in the high-quality development of education, with its influence being contingent on the regional economic level. Furthermore, capital misallocation serves as a conduit between the two factors.

1. Introduction

The role of digital finance in propelling the economy of China has attained growing prominence, owing to its rapid development. In 2023, the digital finance market in China reached a size of RMB 41.7 trillion, accounting for 15.6 % of the global digital finance market and ranking first in the world. Education has been consistently placed at the forefront by the government to ensure the prosperity of the nation. The Fifth Plenary Session of the 19th CPC Central Committee has clearly defined the key requirement of “building a high-quality education system”, to insist on giving priority to the development of education. The integration of digital finance and education can facilitate the optimal distribution of educational resources, enhance educational equity and improve educational quality. Therefore, it is imperative to ascertain how digital finance can be effectively integrated with education to facilitate the high-quality development of education.

Digital inclusive finance is a product of the high degree of integration of financial technology and inclusive finance. This integration determines that digital inclusive finance has both inclusive and digital attributes. The development of digital finance has profoundly impacted the traditional financial industry, significantly enhancing the security and diversity of banking services and effectively improving the efficiency of financial services (Dong et al., 2020; Zhang, 2021). But at the same time, it has a negative impact on bank liquidity creation. For real firms, digital finance shows different degrees of facilitating effects in alleviating corporate financing constraints, increasing investment diversification and reducing disposal effects (Lu et al., 2024), and incentivizing firms' independent innovation and R&D investment (Acharya and Xu, 2016; Liang and Zhang, 2019). Through empirical research on survey data of Chinese rural residents, Ren and Li (2019) found that digital inclusive finance also improves rural inclusive growth. It not only helps to

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improve the entrepreneurial environment, but also brings about the fairness and equalisation of entrepreneurial opportunities (Zhang et al., 2019). The development of digital finance in China has been rapid in economically backward regions, it has led to a significant increase in residents' income, promoted consumption upgrading, and reduced the vulnerability of farm households (Peng and Xu, 2021). Furthermore, digital finance has alleviated China's poverty rate and the degree of income inequality, narrowing the income gap between urban and rural areas (Zhang and Wu, 2021).

Research has shown that there is a close relationship between finance and education. Dou and He (2006) found that the development of finance can provide more financial support for education by investing funds into the education industry. Furthermore, the development of finance can increase financial education expenditure and further increase investment in education (Yang et al., 2021), thus enhancing the return to education (Shen et al., 2015). By participating in the household financial market, it reshapes consumption patterns and heterogeneously influences different categories of household consumption, provides children with more educational opportunities and resource choices, breaks down the original intergenerational educational mobility constraints, and enhances educational equity and social mobility (Zhang and Wan, 2019). Empirical studies by established scholars have shown that digital inclusive finance can play a positive role in enhancing family education expectations (Chen et al., 2021), alleviating family liquidity constraints and preventive savings incentives (Yan et al., 2023), and changing marginal propensity to consume to increase family consumption, which in turn promotes the progress of educational equity (Li et al., 2022).

The survey contributions present in this paper are as follows: Theoretically, we further analyze the impact of digital finance on the high-quality development of education, elucidate its mechanism of action, summarize the law of heterogeneity of digital finance on the high-quality development of education, and enrich the summarized research on the factors influencing the high-quality development of education; practically, the findings of the study can provide useful insights for accurately grasping the opportunities for economic transformation brought about by digital finance, and promoting the high-quality development of education.

This article selects relevant indicators to construct an index of high-quality development of education and sets up an empirical model. The findings indicate that digital finance can significantly promote the high-quality development of local education and realize the promotion effect by alleviating the degree of regional resource mismatch. In areas with insufficient financial development, the positive promotion of digital financial development on the high-quality development of education is more significant. The remainder of the paper is structured as follows: the second part is a theoretical analysis and research hypothesis; the third part is a research design; the fourth part is an empirical analysis, which discusses the impact of digital finance on the upgrading of the education industry, and at the same time explores the role of the regional resource mismatch in this impact mechanism; and the fifth part is a research conclusion and recommendations.

2. Theoretical analysis and research hypotheses

2.1. Digital finance and high-quality development of education

The primary obstacles preventing the economically disadvantaged from pursuing higher education are evident at the pre-school and basic education levels and throughout the education system (Wang et al., 2011). From a micro perspective, the gap between rich and poor at the household level leads to discrepancies in the investment in their children's education and ultimately contributes to the phenomenon of class-based educational poverty (Xie and Shi, 2019). From a macroeconomic perspective, the level of economic development in a region has a significant impact on the education budget. Furthermore, there is a strong correlation between the level of economic development and the level of educational development. On this basis, finance plays a pivotal role in the redistribution of funds through the absorption of deposits, the granting of loans, and the investment of capital. On the one hand, finance provides direct financial support to regions with a shortage of education funds. On the other hand, finance can provide a driving force for the economic growth of backward regions, which in turn can improve the level of economic development and lay the foundation for the abundance of regional education funds. Therefore, finance not only furnishes the requisite material foundation for education, but also enhances the overall quality and efficacy of education and fosters the advancement of education and social advancement.

As a representative of a new financial development model, digital finance promotes the popularisation and deepening of financial services, which can serve and support economic growth (Qian et al., 2020). Therefore, the development of digital finance is of great significance in promoting innovation and development in the field of education. First, digital finance serves to effectively bridge the coverage gap of traditional finance. The derivation and development of digital finance has the effect of creating strong spatial spillovers, which serve to improve the efficiency of capital use and allocation in terms of scale, speed and accuracy (Xu et al., 2022). It uses fintech tools to reduce transaction costs, increase the penetration and accessibility of financial services, and channel more capital to areas in need of education funding. Second, while the convenience of digital finance is appreciated by residents, they are also exposed to new educational ideas and methods. This creates a need for a higher level of education, which in turn leads to increased investment in education for themselves and their families. Furthermore, it promotes the development of the education industry. Digital finance also provides diversified financing channels for the education industry, further accelerating the innovation and application of education technology. With the support of digital finance, financing for the education industry has become more flexible and is no longer limited to traditional bank loans and state subsidies, but can be accessed through platform financing and crowdfunding. This provides a more reliable foundation for the innovation and application of education technology, thereby enhancing the competitiveness and innovation of the education industry.

Based on the preceding analyses, we propose the following hypothesis:

H1. Digital finance, as an important direction of financial development, has a significant positive effect on promoting high-quality

development of education.

2.2. Digital finance, regional resource misallocation and high-quality development of education

The financial system performs two core functions: mobilizing idle funds and identifying investment opportunities. These functions can significantly improve the efficiency of capital allocation, promote the rapid accumulation of capital and stimulate technological innovation (King and Levine, 1993). Wurgler (2000) constructed a capital allocation efficiency assessment model to demonstrate that the enhancement of financial development has a significant positive effect on the efficiency of capital allocation. Digital financial development, as an important component of financial innovation, serves as a significant driver of financial innovation and facilitates the dissemination and optimization of financial services. Firstly, digital finance can effectively correct the "misallocation" in traditional finance by optimising the allocation of resources. Furthermore, in areas with poor financial development, digital finance can demonstrate more inclusive characteristics (Tang et al., 2020). Additionally, digital finance can facilitate the complementary match of resources within a region. The incorporation of big data and other technologies enables digital finance to more accurately identify the resource gap between regions. This allows for the targeted flow of funds to the more resource-poor regions, thereby achieving a balanced distribution of resources. Such a complementary distribution narrows the development gap between regions and enhances the stability and resilience of overall development.

There is a close relationship between education and resource allocation. As a social and public endeavor, the development of education requires the investment of large amounts of financial and other resources. The way these resources are allocated directly determines the scale, structure and operational efficiency of the education system. The rational allocation of resources also has an important impact on educational equity. Xiao et al. (2014) analyzed the inequality of compulsory education in China, identifying the core problem as the allocation of resources. Issues such as the localization of financing make the inequality of educational expenditure persistent and intensified. Reasonable resource allocation also helps to improve the quality and efficiency of education. By optimizing resource allocation and aligning it with the actual requirements of educational development, the effective utilisation of educational resources can be enhanced, and the direction of resource inputs can be adjusted promptly to promote the high-quality development of education. Therefore, Fan (2019) posits that the promotion of social equity and the resolution of resource misallocation and imbalance should be the focus of addressing educational equity.

We propose the following hypothesis based on the preceding analyses:

H2. Digital finance can facilitate the optimal allocation of resources within the region, thereby reducing the degree of resource mismatch and promoting the high-quality development of education.

3. Research design

3.1. Model setting and interpretation of variables

This paper focuses on exploring the impact of regional digital financial development on the high-quality development of education. To test hypothesis 1, this paper constructs model (1):

$$Edu_{i,t} = \alpha_0 + \alpha_1 Finind_{i,t} + \alpha_2 Controls_{i,t} + \mu_i + \lambda_t + \varepsilon_{i,t} \quad (1)$$

3.2. Sample selection and data sources

In this paper, the data at the provincial level in China from 2011 to 2022 are selected as the research sample, and the data are processed and analysed using Stata 16. Considering the accessibility of data, this paper excludes the observations of Hong Kong, Macao, Taiwan and Tibet. The data for the variables in this paper are obtained from China Statistical Yearbook, China Tertiary Industry Statistical Yearbook, China Education Statistical Yearbook, etc. Considering the influence of extreme value data regression, 1 % winsorizing is applied to both ends of continuous variables.

Table 1
Descriptive statistics.

VarName	Obs	Mean	SD	Min	Median	Max
Edu	360	0.270	0.097	0.086	0.266	0.544
Finind	360	−0.001	0.998	−2.077	0.112	1.872
Trans	360	−0.002	0.993	−1.818	−0.074	2.827
GDP	360	0.003	0.967	−2.345	0.093	2.174
Tertiary	360	−0.001	0.980	−2.841	0.227	1.822
Urban	360	0.601	0.120	0.373	0.588	0.893
Popden	360	0.930	0.517	0.133	0.841	2.377
Fintech	360	0.129	0.241	0.003	0.048	1.419
Net	360	−0.003	0.987	−1.155	−0.265	3.357
CMI	360	0.379	0.221	0.017	0.375	0.893

3.3. Descriptive statistics

Table 1 Shows the descriptive statistics for each variable. Specific variable definitions are provided in Appendix A. The mean value of *Edu* is 0.270, the minimum value is 0.086 and the maximum value is 0.544, which presents that there is a large gap in the development of China's education industry. The maximum value of *Finind* is 1.872, the minimum value is -2.077 , and the median is higher than the mean, which shows that the digital financial development of most of China's regions is higher than the average level. The minimum value of *CMI* is 0.017, the maximum value is 0.893, and the standard deviation is 0.221, which indicates that there is a certain degree of variability and misallocation in the resource of capital in China.

4. Empirical tests

4.1. Benchmark regression analysis

Column (1) of Table 2 reports the impact of digital finance on the high-quality development of education. The regression coefficient of the core independent variable *Finind* is 0.194 and is significant at the 1 % level after adding all control variables and fixing the region and time effects. This indicates that digital finance significantly contributes to the high-quality development of education in the region. Hypothesis 1 is verified.

4.2. Robustness tests

4.2.1. Endogeneity test

It has been previously posited that the degree of digital funding in the region exerts a considerable influence on the advancement of high-quality education. Conversely, high-quality education would further promote the innovation of regional digital finance, and the two may be mutually causal. To effectively address the endogeneity issue, this paper employs the number of regional Internet access ports as an instrumental variable to be tested. Firstly, the current financial innovation is closely related to the Internet, big data, blockchain and other technologies, and such technologies are based on the level of regional Internet development. Consequently, there is a correlation between the two. Second, the advancement of education is contingent upon the distribution of regional educational resources and the level of regional educational literacy and ability, given the constraints of Internet technology in the field of teaching and learning. Therefore, the instrumental variable satisfies the exclusivity constraint and does not affect the high-quality development of education through other omitted variables. The correlation is tested statistically. Columns (2) and (3) of Table 2 present the regression results after employing the number of Internet access ports as an instrumental variable. The coefficient of *Finind* is 0.098, which represents a significant positive correlation at the 1 % level. The F-statistic for the weak instrumental variable test is 85.640,

Table 2
Digital finance and high-quality development of education.

	(1) Edu	(2) IV-First	(3) IV-Second	(4) Edu	(5) Edu
Finind	0.194*** (9.43)		0.098*** (5.69)		
Net		0.400*** (9.25)			
Fintech				0.049*** (2.92)	
l.Finind					0.179*** (8.07)
Trans	0.013 (0.97)	0.546*** (3.80)	−0.041 (−1.63)	0.024 (1.59)	0.028* (1.77)
GDP	−0.004 (−0.92)	−0.284*** (−7.31)	0.040*** (5.31)	−0.004 (−0.84)	−0.005 (−1.10)
Tertiary	0.004 (0.84)	0.145*** (4.28)	−0.027*** (−4.55)	0.007 (1.55)	0.006 (1.27)
Patent	−0.014** (−2.08)	0.722*** (13.22)	−0.068*** (−4.83)	−0.025*** (−3.37)	−0.010 (−1.39)
Urban	1.092*** (5.92)	9.129*** (4.76)	−0.593 (−1.50)	1.306*** (5.44)	1.407*** (6.55)
Popden	0.015 (1.48)	−0.197* (−1.77)	0.019 (1.15)	0.006 (0.54)	0.020* (1.80)
_cons	−0.258* (−1.95)			−0.271* (−1.68)	−0.472*** (−3.10)
Year fixed effect	Yes	Yes		Yes	Yes
Province fixed effect	Yes	Yes		Yes	Yes
adj. R^2	0.940		0.128	0.925	0.942
N	360	360	360	360	330

Note: *, **, and *** indicate significance at the 10 %, 5 %, and 1 % levels, respectively.

which is considerably greater than 10 and thus rejects the original hypothesis that the instrumental variable is a weak instrumental variable. Furthermore, the coefficient of *Finind* remains significant after the use of instrumental variables, which simultaneously proves the robustness of the results.

4.2.2. Other robustness tests

(1) Substitution of explanatory variables. According to Xu and Feng (2022), fintech firms are important contributors to the development of digital finance. Therefore, this paper uses regional fintech firm density (number of regional fintech firms/total population of the region) as an alternative way to measure the development of digital finance. (2) Lagged one-period regression. Given the potential for a lag in the impact of digital finance on improving the high-quality development of education, the independent variable *Finind* is lagged by one period, and the main effect is regressed again. The outcomes of the aforementioned robustness tests are presented in columns (4) and (5) of Table 2, respectively. The regression results substantiate the reliability of the benchmark regression conclusions.

5. Mechanism analysis

The phenomenon of capital misallocation refers to the fact that capital is not effectively allocated to the sector that can most enhance productivity. This results in an imbalance in the distribution of educational resources. The development of digital finance, especially inclusive finance, will alleviate the uneven distribution of capital. This paper takes the degree of capital mismatch as a mediating variable to test and constructs the following model.

$$CMI_{i,t} = \alpha_0 + \alpha_1 Finind_{i,t} + \alpha_2 Controls_{i,t} + \mu_i + \lambda_t + \varepsilon_{i,t} \quad (2)$$

$$Edu_{i,t} = \alpha_0 + \alpha_1 Finind_{i,t} + \alpha_2 CMI_{i,t} + \alpha_3 Controls_{i,t} + \mu_i + \lambda_t + \varepsilon_{i,t} \quad (3)$$

The regression results are presented in Table 3. Column (1) is the benchmark model of digital finance affecting the high-quality development of education. Column (2) tests whether there is an effect of digital finance on the extent of capital misallocation, and the results demonstrate that digital finance can inhibit regional capital misallocation. Column (3) incorporates the mediating variable capital misallocation into the benchmark model and finds that capital misallocation has a negative effect on improving education development; the coefficient of digital finance decreases but is still significant. The Sobel test is also statistically significant, and the mediating effect accounted for 8.83 %. In conclusion, the impact of digital finance on the development of high-quality education is mediated by capital mismatch. Hypothesis 2 is therefore tested.

6. Heterogeneity analysis

Given the differences in the level of financial development in the regions, which will have a different impact on the effect of digital

Table 3
Mechanism test results.

	(1) Edu	(2) CMI	(3) Edu
Finind	0.194*** (9.43)	-0.187*** (-2.77)	0.182*** (8.92)
CMI			-0.067*** (-3.99)
Trans	0.013 (0.97)	0.013 (0.30)	0.014 (1.06)
GDP	-0.004 (-0.92)	-0.006 (-0.38)	-0.005 (-1.02)
Tertiary	0.004 (0.84)	0.000 (0.03)	0.004 (0.86)
Patent	-0.014** (-2.08)	0.032 (1.47)	-0.012* (-1.79)
Urban	1.092*** (5.92)	0.051 (0.08)	1.095*** (6.08)
Popden	0.015 (1.48)	-0.113*** (-3.49)	0.007 (0.71)
_cons	-0.258* (-1.95)	0.125 (0.29)	-0.249* (-1.93)
Year fixed effect	Yes	Yes	Yes
Province fixed effect	Yes	Yes	Yes
Sobel Z	2.304**		
Proportion of total effect	8.83 %		
adj. R ²	0.940	0.878	0.943
N	360	360	360

Note: *, **, and *** indicate significance at the 10 %, 5 %, and 1 % levels, respectively.

finance, this paper uses the ratio of regional total deposits and loans to regional GDP as an indicator to measure the level of development of regional finance. Subsequently, based on the median of this indicator, the sample is divided into two groups of high financial development degree and low financial development degree, and regression analyses are conducted separately. The regression results are shown in Table 4.

The regression results presented in columns (1) and (2) in Table 4 indicate that the coefficients are 0.194 and 0.083, respectively, in regions with higher and lower levels of financial development, and both pass the statistical test at the 1 % significance level. This result indicates that the development of digital finance has a significant positive contribution to the high-quality development of education in regions with different economic levels. In addition, the coefficient of the test of difference between groups is -0.023 and is statistically significant. This result shows that the marginal contribution of digital finance is more significant in regions with underdeveloped economic levels. This may be attributed to the fact that the advent and proliferation of digital finance can more effectively address the deficiencies of traditional financial services, thereby assuming a more prominent role in facilitating access to and distribution of educational resources in regions with a relatively less developed economic landscape.

7. Conclusions and recommendations

An empirical analysis of panel data from 30 provinces in China from 2011 to 2022 revealed that digital finance, a crucial aspect of financial development, has a substantial positive impact on the high-quality development of education. This impact is contingent upon the degree of regional resource misallocation. Specifically, the advancement of digital finance can mitigate the degree of regional resource misallocation, thereby further enhancing the improvement of education quality. Moreover, the impact of digital finance on the high-quality development of education is contingent upon the degree of regional financial development. The marginal contribution of digital finance to the high-quality development of education is higher in regions where financial development is inadequate.

Based on the above findings, we present recommendations to different stakeholders.

At the government regulatory level: First, increase support for digital finance in the education sector. It should formulate relevant special policies such as the Action Plan for the High-Quality Development of Digital Finance in Support of Education to encourage it to carry out more innovation in financial products and applications for the education sector; second, optimise resource allocation and increase investment in education. The government can increase its investment in education through digital financial means, such as education funds and education bonds; third, enhance the universality of digital finance and promote educational equity. The government should focus on the synergistic development of eastern and western regions to promote education equity.

At the level of the financial industry: First, provide financial support to carry out education financing; second, promote the innovative application of education technology. Such as intelligent teaching systems, online learning platforms, etc., to improve the efficiency and quality of teaching and learning; Third, to promote fairness and universality of education. For example, special education loans and scholarships are provided to support disadvantaged groups, such as poor students and students with disabilities, to receive quality education and promote educational equity.

Table 4
Heterogeneity analysis results.

	(1) High financial development degree	(2) Low financial development degree
Finind	0.194*** (9.43)	0.083*** (2.88)
Trans	0.013 (0.97)	-0.005 (-0.31)
GDP	-0.004 (-0.92)	0.014** (2.17)
Tertiary	0.004 (0.84)	0.004 (0.73)
Patent	-0.014** (-2.08)	-0.039*** (-4.30)
Urban	1.092*** (5.92)	0.214 (0.54)
Popden	0.015 (1.48)	-0.001 (-0.13)
_cons	-0.258* (-1.95)	0.607** (2.24)
Year fixed effect	Yes	Yes
Province fixed effect	Yes	Yes
Between-group variation test	-0.023***	
adj. R^2	0.940	0.963
N	184	176

Note: *, **, and *** indicate significance at the 10 %, 5 %, and 1 % levels, respectively.

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Appendix

Appendix A. Variable Definitions

A.1. Explained variable: education industry development index (Edu)

After considering the actual situation and the accessibility of data, this paper selects three aspects to measure the development level of the education industry: education investment, education scale and education achievement. The entropy method is employed to calculate the weights of each index and the comprehensive score. Finally, the annual regional education industry development index is obtained.

Table A

System of high-quality development of education.

Primary indicator	Secondary indicator	Calculation method	Nature	Weighting
Investment in education	Investment in fixed assets in education	Annual investment in fixed assets of education industry	+	0.276
	Level of fiscal expenditure on education	Local financial expenditure on education/general budget expenditure	+	0.060
Scale of education	Number of higher education schools	Number of schools at higher education level	+	0.210
	Number of basic education schools	Number of schools below higher education level	+	0.128
	Student-teacher ratio in higher education	Number of students in higher education/number of teachers in higher education	+	0.038
	Student-teacher ratio in basic education	Number of students in basic education/number of teachers in basic education	+	0.084
Educational Achievement	Average years of education	(Number of persons with primary education $\times$ 6 + number of persons with lower secondary education $\times$ 9 + number of persons with upper secondary education $\times$ 12 + number of persons with tertiary education and above $\times$ 16)/population aged 6 and over	+	0.046
	Educational structure	Number of persons with tertiary education/population aged 6 and over	+	0.159

A.2. Explanatory variables: Digital finance index (Finind). This paper utilized the latest Digital Inclusive Finance Index, released by Peking University, to assess the development level of digital finance, spanning from 2011 to 2022. The index encompasses a number of aspects, including the breadth of coverage breadth, the depth of use, and the degree of digitization of digital finance. These indicators permit a more objective assessment of the performance of each region in the digital finance sector.

A.3. Mediating variable: Regional capital misallocation index (CMI). This paper employs the methodology of [Bai and Liu \(2018\)](#) to assess the regional capital resource misallocation. After estimating the factor output elasticity of each region, the amount of capital at the time of effective allocation in each region is calculated. This is then compared with the actual amount of capital to obtain the capital misallocation index. In the actual situation, resource misallocation exists in the case of under-investment and over-investment, so this paper takes the absolute value of the index. The greater the absolute value of the index, the more pronounced the degree of resource misallocation. Conversely, the lower the degree of resource misallocation.

A.4. Control variables. Referring to the relevant empirical studies of [Ma \(2018\)](#) and [Wang et al. \(2020\)](#), the following variables are selected as control variables in this paper to exclude the potential effects of economic level, urbanisation rate and industrial structure of different regions on the explanatory variables. *GDP* indicates the level of regional economic development, measured by the regional

gross domestic product; *Urban* indicates the level of regional urbanisation, which is the proportion of the urban population to the total population; *Trans* indicates the degree of regional transport development, which is equal to the ratio of the local classified highway mileage to the area of the administrative region; *Tertiary* indicates the speed of development of the tertiary industry in the region, which is the added value of the tertiary industry index over the previous year; *Patent* indicates the degree of innovation in the region, measured by the number of patent applications granted in the region; *Popden* is the population density of the region, equal to the number of the population divided by the area of the administrative region. To improve the comparability and credibility of the data, some of the variables are standardised and logarithmic. Additionally, the paper controls for time and region fixed effects to address the endogeneity of the panel data.

Table A

Data availability

The authors are unable or have chosen not to specify which data has been used.

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